1. Write a program in Python
   - To accept 8 numbers, store them in a list NUMS
   - To add adjacent pair of elements of NUMS and store it in another list MNUMS
   - To copy the content of MNUMS on alternate places in the NUMS starting from second place of NUMS.
   - To display content of NUMS as well as MNUMS

   Example:
   ```
   8
   7
   6
   5
   4
   3
   2
   1
   Original Content [NUMS] -> [8, 7, 6, 5, 4, 3, 2, 1]
   8 + 7 = 15
   6 + 5 = 11
   4 + 3 = 7
   2 + 1 = 3
   [NUMS] -> [8, 15, 6, 11, 4, 7, 2, 3]
   [MNUMS] -> [15, 11, 7, 3]
   ```

2. Given a tuple **ALL = (5, 8, 2, 'apple', 'banana', 'Grapes')**, Write a Python code
   I. to print the second to fourth elements of the tuple **ALL**.
   II. to print the content of **ALL** with its content reversed.

3. Consider the following tuples
   ```
   Main = ('Roti', 'Sabji', 'Dal')
   Addon = ('Papad', 'Salad')
   ```
   Write a Python code
   I. to concatenate content of both the tuples and create another tuple **Meal**.
   II. to display the content of **Meal**.

4. Write a Python code to unpack the content of tuple **('Neeraj', 'A-8, ABC Nagar', 2500, 'Male', True)** and store the results in variables **Name**, **Address**, **Fee**, **Gender** and **Indian**. Display the unpacked content from the variables.

5. Given the tuple **Scores = (1, 2, 3, 4, 2, 5, 2, 3, 4, 2, 1)**. Write a Python code to find the count of occurrences of each score from the tuple and display the result in the following format.
   ```
   Score Frequency
   1     2
   2     4
   3     2
   4     2
   5     1
   ```

6. Write a program in Python
   - To accept item numbers of 5 items, store them in a list INO
   - To accept item names of 5 items, store them in a list INAME
   - To create a dictionary ITEM with Key-Value pair with Keys from INO and corresponding Values from INAME
   - To display the content of INO and INAME
   - To display the content of ITEM in ascending order of item numbers

7. **A. Assign the following contents in a tuple Names**
   "JAYA", "AMAR", "PRIYA", "AKBAR", "RESHMA", "ANTHONY"

B. Assign the following contents in a tuple Marks
`75, 56, 86, 92, 65, 86`

- Create a dictionary **Results** with keys from tuple **Names** and corresponding values from tuple **Marks**.
- Display the content of **Results** with keys arranged alphabetically in ascending order.
- Display the content of values of **Results** arranged in ascending order.
- Create a new dictionary named **Toppers** to store only such items of dictionary **Results** where marks are more than **80**.

8. Write a Python code to perform the following:
- To accept 8 numbers in a loop, store them in a Tuple T (with the help of re-assignment method)
- To display the content of T in reverse order
- To add and display the sum of values stored in T
- To find and display minimum and maximum values present in T
- To display sum of each adjacent pair of values
- To find those pairs (any pair 1st-2nd, 1st-5th, 3rd-6th,...) of values from the content of T, whose sum is the same as one of the values in the tuple T.

9. Write a Python code to perform the following:
- To initialize a tuple WD containing **(1,2,3,4,5,6,7)**
- To initialize a list WDN containing **['SUN','MON','TUE','WED','THU','FRI','SAT']**
- To create a dictionary W with key-value pairs with corresponding values from WD and WDN
- To display content of W
- To re-arrange the content of dictionary in such a way that it becomes as follows: **{1:'MON',2:'TUE',3:'WED',4:'THU,5:'FRI',6:'SAT',7:'SUN'}**
- To display the content of W
- To copy the partial content of W in dictionaries **MyDays** and **OfficeDays**, **MyDays** should have content from keys 2,4 and 7 and rest from W to become the content of **OfficeDays**.
- To display the contents of **MyDays** and **OfficeDays**

10. Write a Python code to perform the following operations:
- To initialize a tuple **TL=('RED','YELLOW','GREEN')**
- To accepts names of **10** colors from user and store them in a list **CL**
- To display the color names from **CL** along with corresponding message **"TRAFFIC LIGHT"** and **"NOT TRAFFIC LIGHT"** after checking from the content of **TL**
- To initialize another tuple **TM=('STOP','BE READY TO START/STOP','GO')**
- To create a dictionary **TLM** by combining corresponding key-value pairs from **TL** and **TM**.
- To display the content of **TLM**

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN THE ASSIGNMENT BOOKLET

General Instructions:

i. Type and execute the solutions of the above mentioned problems on IDLE/colab

ii. Type the following on top of your program code with desired information about each of your programs as comment line (in the same format) - It is mandatory to use **Courier New/Fixed Size Font** with **Style – BOLD & Size – 11** or **12** in all the programs and also **use single line spacing** to avoid wastage of papers:

```
'''Program No      : 01 (Practical List 5)
   Developed By    : Aarya Singhraj
   Class Section   : XI H
   Date            : 09-Dec-2024'''
```

iii. On successful execution, copy and paste the sample output at the bottom of the program as comment lines. Give program filename as per list and practical number as L5P1.PY, L5P2.PY,...

iv. Save all the programs in a google folder with a name as <Section>-<YourName>-CS (Example: XI-H-RAMESH-CS) and share with CS Teacher. Take a hard copy (printout) of the program and get it signed from the respective CS teacher along with an index entry in a Practical File.

```
# ------------------------------------------------------------#
# List-Program No   : L5-P1
# Developed By      : Puja Dey
# Date              : 29-November-2024
# ------------------------------------------------------------#

NUMS=[]
for i in range(1,9,1):
  N=int(input("Enter a number: "))
  NUMS.append(N)
print("Original Content [NUMS] -> ", NUMS)
MNUMS=[]
for j in range (0,8,2):
  s=NUMS[j]+NUMS[j+1]
  print(NUMS[j],"+", NUMS[j+1],"=",s)
  MNUMS.append(s)
for k in range(0,4):
  NUMS[2*k+1]=MNUMS[k]
print("[NUMS] -> ", NUMS)
print("[MNUMS] -> ", MNUMS)
```

```
Enter a number: 8
Enter a number: 7
Enter a number: 6
Enter a number: 5
Enter a number: 4
Enter a number: 3
Enter a number: 2
Enter a number: 1
Original Content [NUMS] ->  [8, 7, 6, 5, 4, 3, 2, 1]
8 + 7 = 15
6 + 5 = 11
4 + 3 = 7
2 + 1 = 3
[NUMS] ->  [8, 15, 6, 11, 4, 7, 2, 3]
[MNUMS] ->  [15, 11, 7, 3]
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P2
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#

ALL = (5, 8, 2, 'apple', 'banana', 'Grapes')
print(ALL[1:4])
print(ALL[::-1])
```

```
(8, 2, 'apple')
('Grapes', 'banana', 'apple', 2, 8, 5)
```

```
# ------------------------------------------------------------------#
# List-Program No   : L5-P3
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------------#

Main = ('Roti', 'Sabji', 'Dal')
Addon = ('Papad', 'Salad')
Meal=Main+Addon
print(Meal)
('Roti', 'Sabji', 'Dal', 'Papad', 'Salad')
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P4
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#
INFO=('Neeraj', 'A-8, ABC Nagar', 2500, 'Male', True)
Name, Address, Fee, Gender, Indian = INFO
print("Name:", Name)
print("Address:", Address)
print("Fee:", Fee)
print("Gender:", Gender)
print("Indian:", Indian)
```

```
Name: Neeraj
Address: A-8, ABC Nagar
Fee: 2500
Gender: Male
Indian: True
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P5
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#
Scores = (1, 2, 3, 4, 2, 5, 2, 3, 4, 2, 1)
print("Scores", "Frequency", sep=' ')
for i in range(1,6):
  Fi=Scores.count(i)
  print(i,Fi, sep='      ')
```

```
Scores Frequency
1      2
2      4
3      2
4      2
5      1
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P6
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#
INO=[]
for i in range(1,6):
  C=input("Enter the ITEM NO: ")
  INO.append(C)
print(INO)

INAME=[]
for i in range(1,6):
  C=input("Enter the ITEM NAME: ")
  INAME.append(C)
print(INAME)

ITEM={}
for i in range(0,5):
  ITEM[INO[i]]=INAME[i]
print(sorted(ITEM.items()))
```

```
Enter the ITEM NO: 1
Enter the ITEM NO: 5
Enter the ITEM NO: 2
Enter the ITEM NO: 4
Enter the ITEM NO: 3
['1', '5', '2', '4', '3']
Enter the ITEM NAME: ONE
Enter the ITEM NAME: FIVE
Enter the ITEM NAME: TWO
Enter the ITEM NAME: FOUR
Enter the ITEM NAME: THREE
['ONE', 'FIVE', 'TWO', 'FOUR', 'THREE']
[('1', 'ONE'), ('2', 'TWO'), ('3', 'THREE'), ('4', 'FOUR'), ('5', 'FIVE')]
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P7
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#
Results={}
Names=("JAYA", "AMAR", "PRIYA", "AKBAR", "RESHMA", "ANTHONY")
Marks=(75, 56, 86, 92, 65, 86)
for i in range(len(Marks)):
  Results[Names[i]]=Marks[i]
print('Results:', Results)
A=sorted(Results.keys())
B=sorted(Results.values())
print('Keys', A)
print('Values', B)
```

```
Results: {'JAYA': 75, 'AMAR': 56, 'PRIYA': 86, 'AKBAR': 92, 'RESHMA': 65, 'ANTHONY':
    86}
Keys ['AKBAR', 'AMAR', 'ANTHONY', 'JAYA', 'PRIYA', 'RESHMA']
Values [56, 65, 75, 86, 86, 92]
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P8
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#
T = ()
for i in range(8):
    num = float(input("Enter number: "))
    T = T + (num,)
print()
print("Original: ", T)
print("Reverse: ", T[::-1])
print()
s=0
for j in range(8):
  s=s+T[j]
print("Sum of all terms: ", s)
print()
print("Min", min(T))
print("Max", max(T))
print()
ss=0
print("Sum of Adjacent Pairs")
for k in range (0,8,2):
  ss=T[k]+T[k+1]
  print(T[k],"+", T[k+1],"=",ss)
print()
pairs=[]
for m in range(len(T)):
    for n in range(m + 1, len(T)):
        pair_sum = T[m] + T[n]
        if pair_sum in T:
           pairs.append((T[m], T[n], pair_sum))
for x in pairs:
    print("Pair:", x[0], "+", x[1], "=", x[2])
```

```
Enter number: 8
Enter number: 7
Enter number: 6
Enter number: 5
Enter number: 4
Enter number: 3
Enter number: 2
Enter number: 1

Original:  (8.0, 7.0, 6.0, 5.0, 4.0, 3.0, 2.0, 1.0)
Reverse:  (1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0)
```

```
Sum of all terms:  36.0

Min 1.0
Max 8.0

Sum of Adjacent Pairs
8.0 + 7.0 = 15.0
6.0 + 5.0 = 11.0
4.0 + 3.0 = 7.0
2.0 + 1.0 = 3.0

Pair: 7.0 + 1.0 = 8.0
Pair: 6.0 + 2.0 = 8.0
Pair: 6.0 + 1.0 = 7.0
Pair: 5.0 + 3.0 = 8.0
Pair: 5.0 + 2.0 = 7.0
Pair: 5.0 + 1.0 = 6.0
Pair: 4.0 + 3.0 = 7.0
Pair: 4.0 + 2.0 = 6.0
Pair: 4.0 + 1.0 = 5.0
Pair: 3.0 + 2.0 = 5.0
Pair: 3.0 + 1.0 = 4.0
Pair: 2.0 + 1.0 = 3.0
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P9
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#
WD = (1,2,3,4,5,6,7)
WDN = ['SUN','MON','TUE','WED','THU','FRI','SAT']
W={}
for i in range(0,7):
  W[WD[i]]=WDN[i]
print(W)

values = list(W.values())
for j in range(1, 7):
    W[j] = values[j]
W[7] = values[0]
print(W)
MyDays={W[2],W[4],W[7]}
print("MyDays: ", MyDays)
OfficeDays={W[1], W[3],W[5],W[6]}
print("OfficeDays: ", OfficeDays)
```

```
{1: 'SUN', 2: 'MON', 3: 'TUE', 4: 'WED', 5: 'THU', 6: 'FRI', 7: 'SAT'}
{1: 'MON', 2: 'TUE', 3: 'WED', 4: 'THU', 5: 'FRI', 6: 'SAT', 7: 'SUN'}
MyDays:  {'SUN', 'THU', 'TUE'}
OfficeDays:  {'WED', 'SAT', 'FRI', 'MON'}
```

```
# ------------------------------------------------------------#
# List-Program No   : L5-P10
# Developed By    : Puja Dey
# Date            : 29-November-2024
# ------------------------------------------------------------#
CL=[]
TL=('RED','YELLOW','GREEN')
for i in range(1,11):
  C=input("Enter the name of colour: ")
  CL.append(C)
print(CL)

for i in range(len(CL)):
  if CL[i] in TL:
    print(CL[i], ":A Traffic Light")
  else:
    print(CL[i], ":Not A Traffic Light")

TLM={}
TM=('STOP','BE READY TO START/STOP','GO')
for i in range(0,3):
  TLM[TL[i]]=TM[i]
print(TLM)
```

```
Enter the name of colour: RED
Enter the name of colour: YELLOW
Enter the name of colour: BLUE
Enter the name of colour: GREEN
Enter the name of colour: ORANGE
Enter the name of colour: GREY
Enter the name of colour: BLACK
Enter the name of colour: PURPLE
Enter the name of colour: PINK
Enter the name of colour: DARK BLUE
['RED', 'YELLOW', 'BLUE', 'GREEN', 'ORANGE', 'GREY', 'BLACK', 'PURPLE', 'PINK', 'DARK
    BLUE']
RED :A Traffic Light
YELLOW :A Traffic Light
BLUE :Not A Traffic Light
GREEN :A Traffic Light
ORANGE :Not A Traffic Light
GREY :Not A Traffic Light
BLACK :Not A Traffic Light
PURPLE :Not A Traffic Light
PINK :Not A Traffic Light
DARK BLUE :Not A Traffic Light
{'RED': 'STOP', 'YELLOW': 'BE READY TO START/STOP', 'GREEN': 'GO'}
```